Scheme of Valuation

25901919

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Elective(DSE)-MG3DSEPHY202 – INTRODUCTION TO SPACEPHYSICS

(2024 ADMISSION ONWARDS)

Duration:1Hour 30Minutes **Maximum Marks:50**

Remember(K), Understand(U),Apply(A),Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. What is cold plasma? [U] / [CO1]

In cold plasma, an electric field energizes the electrons to a high temperature, but the electrons do not collide with the ions and neutral particles often enough to heat them significantly. This allows the plasma to exist at a much lower overall temperature, sometimes near room temperature.
It is found in Earth's atmosphere, Neon signs and fluorescent light bulbs, Plasma globes

For answer with these points give two marks. For partial answers give marks accordingly

2. Mention two challenges faced in human spaceflight missions. [Ap] / [CO6]

Microgravity (Weightlessness): Prolonged exposure to microgravity causes muscle atrophy, bone density loss, fluid redistribution, and balance problems, affecting astronauts' health.
Space Radiation: Astronauts are exposed to high levels of cosmic rays and solar radiation outside Earth's protective atmosphere and magnetic field, increasing the risk of cancer, tissue damage, and other health issues.

For answer with these points give two marks. For partial answers give marks accordingly

3. Why is studying coronal mass ejections important for Earth's technology systems? [U] / [CO4]

Occasionally the Sun's surface erupts and sends a large part of the solar atmosphere streaming away at high speeds. These events, called coronal ejections (CMEs), can contain 1012 kg of material and can move away from the Sun at over 1000 km /s. If CMEs are directed towards Earth, a great space storm can develop far above our heads, destroying satellites, causing increased radiation exposure for airline crews and passengers, blacking out some forms of radio communication, and disrupting power systems on Earth mass

For answer with these points give two marks. For partial answers give marks accordingly

4. What role does space weather forecasting play in power grid protection? [U] / [CO5]

The solar wind is not steady or uniform but changes constantly. These changes affect Earth's space environment in a number of ways, including large electrical currents that can disrupt communication, power grids, and satellite navigation. If CMEs are directed towards Earth, a great space storm can develop far above our heads, destroying satellites, causing increased radiation exposure for airline crews and passengers, blacking out some forms of radio communication, and disrupting power systems on Earth. The emerging science of space weather is attempting to understand the causes of space storms and their impact on Earth's technological infrastructure with the hope that we can forecast space weather and reduce damage.

For answer with these points give two marks. For partial answers give marks accordingly

5. How do solar energetic particles affect satellites? [U] / [CO2]

In 2020 there were over 3368 operational satellites currently orbiting Earth. Many of these are commercial communications satellites that provide that provide global news TV coverage, telephone connections, and credit card transactions. All of these satellite systems are susceptible to damage and degradation due to the harsh space environment and contribute to orbital debris that increases the chance of operational satellite collisions.

For answer with these points give two marks. For partial answers give marks accordingly

6. What is space weather? [U] / [CO2]

Space weather is the emerging field within the space sciences that studies how the Sun influences Earth's space environment and the technological and societal impacts of that interaction: Damage to or destruction of Earth orbiting satellites, Threats to both astronaut safety during long-duration missions to the Moon and Mars, Threats to the reliability and accuracy of global communications and navigation systems.

For answer with these points give two marks. For partial answers give marks accordingly

7. What is the main cause of solar flares? [K] / [CO2]

A solar flare is a sudden, intense burst of radiation and energy from the surface of the Sun. They occur in active regions—usually around sunspots—where strong magnetic fields twist and release energy

Twisting of Magnetic Field Lines: The Sun's intense plasma movement causes its magnetic field lines near sunspots to twist, stretch, and tangle over time, storing massive amounts of magnetic energy.

Magnetic Reconnection: When these opposite magnetic field lines cross and reorganize, they suddenly snap and reconnect. This process—called magnetic reconnection—rapidly releases the stored magnetic energy, resulting in a solar flare

For answer with these points give two marks. For partial answers give marks accordingly

8. What is the main process behind the energy production in the Sun's core? [K] / [CO4]

$$4\,{}_{1}^{1}\mathrm{H} + 2e^{-} \rightarrow {}_{2}^{4}\mathrm{He} + 2\nu + 6\gamma + 26.7\,\mathrm{MeV}$$

For answer with these points give two marks. For partial answers give marks accordingly

9. State any two applications of the measurement of TEC from GPS data. [K] / [CO5]

The measurement of Total Electron Content (TEC) from GPS data is highly useful for studying the Earth's atmosphere and ensuring the reliability of technology

Improving GPS Navigation and Positioning Accuracy

The ionosphere alters the speed and path of radio signals passing through it, introducing time delays proportional to the TEC. By measuring TEC in real time, dual-frequency GPS receivers can calculate and eliminate these ionospheric delays, while single-frequency systems can use the data to refine empirical ionospheric models, significantly reducing positioning errors.

Space Weather Monitoring and Forecasting

TEC measurements act as an indicator of ionospheric variability caused by solar events, such as solar flares, geomagnetic storms, and coronal mass ejections. Tracking TEC fluctuations allows scientists to monitor space weather conditions that disrupt satellite operations, high-frequency (HF) radio communications, and power grids.

For answer with these points give two marks. For partial answers give marks accordingly

10. Briefly discuss the magnetic reconnection in the magnetosphere of Jupiter. [K] / [CO3]

Magnetic reconnection in Jupiter's magnetosphere is the process where magnetic field lines break and rapidly reconnect, converting stored magnetic energy into plasma kinetic and thermal energy.

1. Dayside Magnetopause (Flux-Transfer Events): Reconnection occurs on Jupiter's dayside magnetopause, manifesting as Jovian flux-transfer events (FTEs) where the planet's magnetic field interacts directly with the solar wind.

2. Magnetotail Dynamics & Material Release: In the Jovian magnetotail, reconnection plays a critical role in plasma transport. As material flows around Jupiter into the tail region, reconnection breaks the closed magnetospheric field lines, allowing the trapped material to escape, flow downstream through the tail, and ultimately vent out into the solar wind.

For answer with these points give two marks. For partial answers give marks accordingly

11. Define sporadic E. [K] / [CO3]

Sporadic E refers to unpredictable, thin clouds of unusually high electron density that form within the E region of Earth's ionosphere. Typically it is at altitudes between 90 km and 130 km.

For answer with these points give two marks. For partial answers give marks accordingly

12. State the relationship of plasma frequency to electron density. [K] / [CO3]

The plasma frequency (f_p) is directly proportional to the square root of the electron density (Ne)

$$f_N = \sqrt{\frac{N_e e^2}{4\pi^2 \varepsilon_0 m}}$$

For answer with these points give two marks. For partial answers give marks accordingly

13. Is exosphere isothermal? Justify your answer with reason. [U] / [CO4]

The exosphere is not isothermal; its temperature varies dramatically depending on solar activity and the time of day. While its temperature becomes nearly constant with increasing altitude above a certain point (the exobase) and tends to average very high values, it is not a uniform temperature throughout the layer itself or over time.

For answer with these points give two marks. For partial answers give marks accordingly

14. Why is a southward IMF orientation important for magnetospheric activity? [U] / [CO5]

A southward Interplanetary Magnetic Field (IMF) orientation points opposite to Earth's intrinsic northward-pointing magnetic field at the dayside magnetopause. This antiparallel alignment is crucial for magnetospheric activity because of two reasons

1. **Triggers Magnetic Reconnection:** It allows efficient magnetic reconnection to occur between the solar wind and Earth's magnetic field on the dayside, breaking and reconnecting field lines to open up the magnetosphere.

2. **Drives Energy Transfer & Storms:** Opened field lines are swept back by the solar wind into the magnetotail (the Dungey Cycle), transferring massive amounts of solar wind energy, mass, and momentum into the system. This powers geomagnetic storms, substorms, enhanced convection, and intense auroral displays.

For answer with these points give two marks. For partial answers give marks accordingly

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. Explain the process of geomagnetic storms caused by solar activity and their impact on Earth. [U] / [CO1]

A geomagnetic storm is a temporary disturbance of Earth's magnetic field caused by intense solar activity.

The Sun emits solar flares and Coronal Mass Ejections (CMEs), releasing enormous amounts of charged particles into space. These particles travel through the solar wind and reach Earth. When the charged particles interact with Earth's magnetosphere, they compress and distort the magnetic field. This interaction generates strong electric currents and causes fluctuations in Earth's magnetic field, resulting in a geomagnetic storm.

Impacts on Earth

- **Communication Disruptions:** High-frequency radio communication and satellite communication may be affected.
- **GPS Errors:** Navigation systems can become less accurate.
- **Power Grid Failures:** Induced electric currents can damage transformers and cause power outages.
- **Satellite Damage:** Increased radiation can interfere with satellite electronics and shorten satellite lifetimes.
- **Risk to Astronauts:** Higher radiation exposure poses health risks to astronauts and high-altitude flights near the poles.

For answer with these points give 5 marks. For partial answers give marks accordingly

16. Describe the growth of space physics from early aurora observations to the discovery of Earth’s Van Allen belts. [An] / [CO6]

The development of space physics began with early observations of natural phenomena in the sky and advanced rapidly with the advent of the space age.

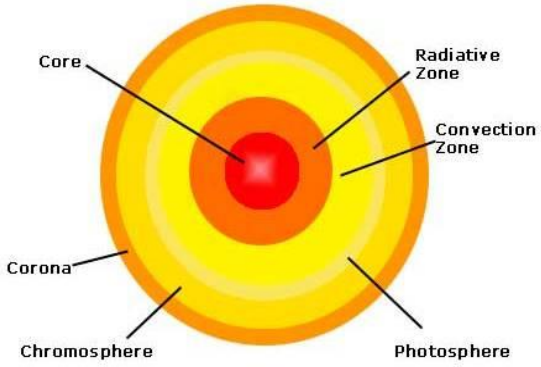
1. Early Aurora Observations: Ancient civilizations observed the aurora borealis and aurora australis but could not explain their origin. In the 19th century, scientists recognized that auroras were linked to Earth's magnetic field and solar activity.
2. Study of Earth's Magnetism: Scientists such as William Gilbert and Carl Friedrich Gauss made significant contributions to understanding Earth's magnetic field, laying the foundation for geomagnetism.

3. Solar–Terrestrial Connection: During the 19th and early 20th centuries, researchers discovered that sunspots**,** solar flares, and the solar wind influence Earth's magnetic field and produce geomagnetic storms and auroras.
4. Beginning of the Space Age: The launch of Sputnik 1 in 1957 marked the beginning of the space age, allowing scientists to study near-Earth space using satellites.
5. Discovery of the Van Allen Belts (1958): Instruments aboard the Explorer 1 satellite, led by James Van Allen, discovered two doughnut-shaped regions of trapped high-energy charged particles surrounding Earth. These are known as the Van Allen radiation belts**.**

The discovery of the Van Allen belts confirmed that near-Earth space is an active plasma environment and marked the birth of modern space physics**,** leading to extensive research on the magnetosphere, solar wind, space weather, and their effects on satellites, astronauts, and communication systems.

For answer with these points give 5 marks. For partial answers give marks accordingly

17. With a neat diagram explain the atmosphere of Sun. [U] / [CO2]



Explanation of photosphere, chromosphere and corona.

For answer with these points give 5 marks. For partial answers give marks accordingly

18. Explain how SOHO's location atL1 benefits continuous solar observation. [U] / [CO4]

L1 here refers to Lagrange Point 1 of the Sun-Earth system. The Lagrangian Point L1 (also written as L1 point) is a special position in space between the Earth and the Sun where the gravitational forces of both bodies and the orbital motion of a satellite balance each other. At this point, a satellite can "stay" in line with the Earth and Sun as both move around in space. The gravitational pull of the Earth and the Sun is balanced by the centrifugal force of the satellite's orbit, so it needs very little fuel to maintain its position.

For answer with these points give 5 marks. For partial answers give marks accordingly

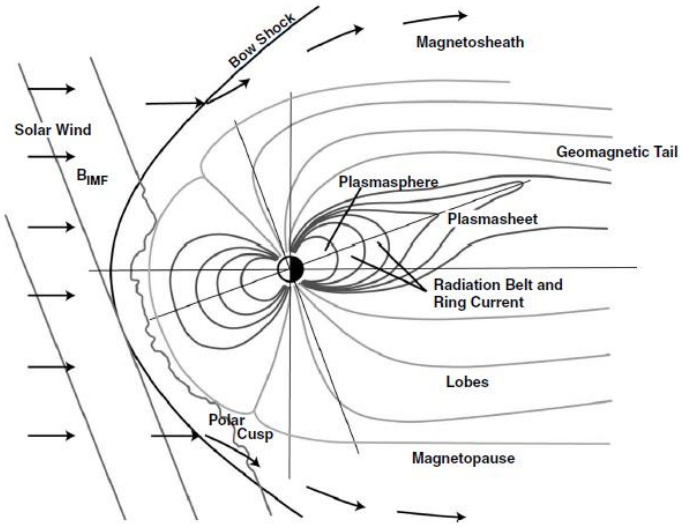
19. Explain the origin of the solar wind and how its properties vary with solar activity. [U] / [CO5]

The solar wind is a stream of charged particles released from the upper atmosphere of the Sun, called the corona. This plasma mostly consists

of electrons, protons and alpha particles with kinetic energy between 0.5 and 10 keV. The composition of the solar wind plasma also includes atomic nuclei C, N, O, Ne, Mg, Si, S, and Fe. There are also rarer traces of some other nuclei and isotopes such as P, Ti, Cr, Ni, Fe 54 and 56, and Ni 58,60,62. The solar wind varies in density, temperature and speed over time and over solar latitude and longitude. Its particles can escape the Sun's gravity because of their high energy resulting from the high temperature of the corona

For answer with these points give 5 marks. For partial answers give marks accordingly

20. With a neat diagram, describe the structure of Earth's magnetosphere. [U] / [CO3]



The magnetosphere is a cavity carved out of the solar wind by Earth's intrinsic magnetic field. It acts as a shield, protecting the planet's atmosphere from direct erosion by the solar wind. The main structural components are listed below.

The complex structure can be broken down into specific regions, structured by the interaction between the incoming solar wind and Earth's magnetic dipole:

1. Bow Shock: This is the initial boundary (like a sonic boom wave) forming far upstream on the dayside where the supersonic solar wind first encounters Earth's magnetic field and is abruptly slowed down to subsonic speeds.
2. Magnetosheath: The region immediately downstream of the bow shock where the compressed, slowed, and heated solar wind plasma flows around the magnetosphere. This plasma is more turbulent.
3. Magnetopause: The fundamental current layer and boundary defining the outer edge of the magnetosphere proper. It is where the outward pressure from Earth's magnetic field exactly balances the inward dynamic pressure of the solar wind.
4. Magnetosphere Proper: The region inside the magnetopause, dominated by Earth's geomagnetic field. It contains several interior regions:
    - Plasmasphere: A relatively dense, cold plasma region originating from Earth's upper ionosphere, typically toroid-shaped and trapped inside the inner magnetosphere (co-rotating

with Earth).
    - Radiation Belts (Van Allen Belts): Regions of highly energetic charged particles (protons and electrons) trapped by Earth's field, shown as intense toroids.
5. Magnetotail: The portion of the magnetosphere stretching far away from the Sun (behind Earth), shaped into a long tail by the solar wind. Key features include:
    - Tail Lobes: Large, low-density regions where magnetic field lines originating from the polar caps are stretched back straight by the solar wind.
    - Plasma Sheet: A dynamic sheet of hotter, denser plasma located in the central equatorial plane between the north and south tail lobes.

Cusps: Funnel-like regions above the dayside magnetic poles where the magnetopause does not form a complete barrier, allowing a small portion of solar wind plasma to directly enter and precipitate down into Earth's ionosphere, potentially fueling auroras.

For answer with these points and figure give 5 marks. For partial answers give marks accordingly

21. Discuss the primary and secondary processes which contribute to the production of charge carriers in ionosphere. [U] / [CO4]

Various processes contribute to the production of charge carriers. The primary process to be considered is photoionization of thermospheric gases by solar EUV and X-ray radiation. Secondary processes include ionization by photoelectrons and scattered or reemitted radiation, but particularly charge exchange reactions. Finally, precipitation of energetic particles plays an essential role in the ionization production at polar latitudes.

Primary Photoionization: Simple (i.e. non dissociative) photoionization processes have the general form

$x + photon (\lambda \sim 100nm) \rightarrow X^+ + e$

where X is an atom or molecule in the upper atmosphere. Of particular importance for the ionosphere is photoionization of the predominant gases 0, N_2 and O_2

$O + photon (\lambda \leq 91\ nm) \rightarrow O^+ + e$

$N_2 + photon (\lambda \leq 80\ nm) \rightarrow N_2^+ + e$

$O_2 + photon (\lambda \leq 103\ nm) \rightarrow O_2^+ + e$

Secondary Ionization Processes: The photoelectrons released as a result of the primary ionization process can attain sufficiently high energies to ionize neutral gas particles themselves. For example, a photon of wavelength ≈30.4 nm can produce a photoelectron from an oxygen atom with a kinetic energy of almost 27 eV, which is larger than the ionization energy of atmospheric gases. Another possibility is that energetic photoelectrons can induce neutral gas particles to emit EUV radiation, there by triggering additional photoionization events. It is estimated that the ionization rate due to these secondary ionization processes, when averaged over all wavelength ranges and constituents, contributes about 20 % of the primary production rate in the upper ionosphere. The secondary production rate in the E region can even be as large as that from the primary photoionization processes.

Charge Exchange. Simple (i.e. nondissociative) charge exchange reactions have the general form

$X + Y^+ \rightarrow X^+ + Y$

Although ions of the species X+ are produced by these reactions, ions of the

species Y^+ are lost, so that the total ionization density remains the same.
Particle Precipitation: In addition to photons, the upper atmosphere can be ionized by incident energetic particles, and this process plays a particularly important role at high latitudes. For precipitating energetic electrons this process may be written as
$X + e_{primary} (E< 12 eV) \rightarrow X^+ + e_{secondary} + e_{prirnary}$
where the index 'primary' denotes the precipitating electron and the index 'secondary' the newly produced secondary electron. In contrast to photons, energetic particles can induce many primary ionization events (auroral electrons have energies from a few 100 up to many 1000 eV), thereby considerably complicating a calculation of the total effect. Furthermore, the secondary electrons are mostly energetic enough to induce ionization events themselves, and this holds as well for any tertiary electrons and all other electrons of higher order.

For answer with these points give 5 marks. For partial answers give marks accordingly

22. Describe the formation of a bow shock in front of a planetary magnetosphere. [U] / [CO5]

A bow shock is a stationary collisionless shock wave formed upstream of a magnetized planet when the solar wind encounters the planet's magnetic field. It serves as the boundary where the solar wind is slowed from supersonic to subsonic speeds, allowing it to divert around the obstacle.

Formation Process

1. Supersonic Solar Wind Stream

   The solar wind is a high-speed stream of ionized plasma (protons and electrons) carrying the Interplanetary Magnetic Field (IMF). As it travels away from the Sun, its velocity exceeds both the local sound speed and the Alfvén speed (the speed at which magnetic waves propagate in a plasma). Thus, the flow is both supersonic and super-Alfvénic (typically Mach 5–10).

2. Obstacle Encounter (The Magnetopause)

   When the solar wind reaches a planet with an intrinsic magnetic field, it cannot freely penetrate Earth's magnetic boundary (the magnetopause). The planet's magnetic pressure acts as an impenetrable barrier to the oncoming plasma.

3. Formation of the Shock Wave

   Because the solar wind is moving faster than the characteristic signal speed (Alfvén speed) of the plasma, pressure disturbances caused by the obstacle cannot travel upstream to "warn" or gradually deflect the incoming flow.

   To adjust to the barrier, the super-Alfvénic solar wind must decelerate abruptly within a very short distance (much smaller than the collisional mean free path of the particles). This steep, non-linear pressure buildup gives rise to a curved bow shock wave ahead of the magnetopause.

4. Plasma Processing Across the Shock Boundary

   As solar wind particles cross the bow shock, kinetic energy is converted into thermal and magnetic energy through collisionless plasma mechanisms (such as wave-particle interactions, micro-instabilities, and magnetic turbulence):

   - Speed: Plummets from supersonic to subsonic.
   - Density & Pressure: Increase sharply (plasma compresses).
   - Temperature: Increases significantly (plasma heats up).
   - Flow direction: Deflects laterally around the magnetopause.

5. Creation of the Magnetosheath

   The decelerated, compressed, and turbulent plasma region immediately downstream of the bow shock is called the magnetosheath. This layer buffers the magnetopause, allowing the solar wind flow to smoothly wrap around the magnetosphere and stream down into the interplanetary medium

For answer with these points give 5 marks. For partial answers give marks accordingly

[5x6 = 30]